

Worked for 38s

A6 — Character Routes

This is where we test whether the Headquarters behaves like a **real organisation with people doing work**, rather than a building that only exists for the player.

A6 is also different from A5:

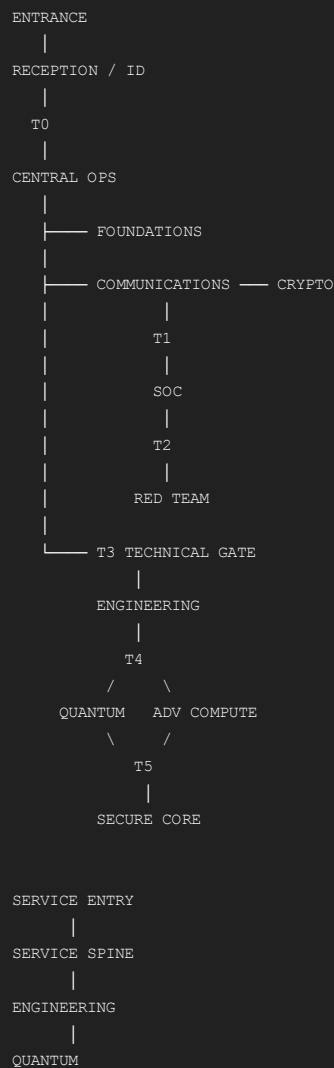
- **A5** asked: *where does the learner move during the campaign?*
- **A6** asks: *where does everybody else belong, how do they move, what access do they have, and do their movements still make sense when the player is not interacting with them?*

The first rule I would lock is:

Characters move because their role gives them a reason to move. They do not follow the learner around simply because the learner needs exposition.

A6.1 — Character route network

The A4 Headquarters already gives us a useful navigation graph:



Every character does **not** have access to every branch.

That becomes the foundation of both:

- NPC pathfinding;
- clearance;
- believable organisational behaviour.

A6.2 — Five types of character movement

I would distinguish these from the beginning.

1. Ambient movement

Normal organisational life.

```
desk
↓
printer / console
↓
another workstation
↓
return
```

It gives HQ life.

2. Operational movement

A character genuinely needs another department.

Example:

```
Alice
Central Ops
↓
Communications
↓
Crypto
```

3. Incident movement

The Headquarters changes because an incident is active.

Example:

```
SOC analyst
SOC
↓
Central Ops liaison point
↓
SOC
```

4. Scripted narrative movement

Specific campaign sequences.

Example:

```
Alice
↓
approaches Workstation 04
↓
hands over USB
```

This must override ordinary ambient behaviour.

5. Remote presence

A character participates without physically entering the room.

The most important example is:

Bob

That distinction will save us from creating ridiculous character routes.

A6.3 — Player route

A5 already gives the player's campaign route, so we do not need to redesign it here.

However, one requirement matters for NPC design:

NPC traffic should never make the player's route feel obstructed.

So player circulation receives priority.

We should reserve approximately:

```
MAIN PLAYER ROUTE
≥ 1.5 m clear usable path

MAIN CORRIDORS
3.0 m

INTERACTION POCKETS
approximately 1.5-2 m clear around important objects
```

NPCs can occupy the environment without constantly blocking doors or Workstation 04.

A6.4 — Alice

Alice is probably the most important physical NPC in the early campaign.

Her natural organisational triangle is:



That is very good because A4 already places these spaces close together.

Alice — normal role

Alice should primarily belong to:

```
CENTRAL OPERATIONS
COMMUNICATIONS
CRYPTOGRAPHY
```

She should **not** routinely appear in:

```
Reception
Red Team
Secure Core
Quantum
```

unless a later mission gives her a genuine reason.

That prevents her from becoming an exposition NPC who magically appears everywhere.

A6.5 — Alice during the Prologue

This reveals one correction to our earlier prototype.

Previously we considered:

```
AliceSpawn  
(0,0,-6)
```

near the southern Central Ops entrance.

I would change that.

The trainee enters Central Operations from the **south** through Reception.

Alice should not appear to enter using that same public trainee route.

Instead:

```
PLAYER  
Reception  
↓  
South Central Ops entrance  
  
ALICE  
Internal operational route  
↓  
North / north-west Central Ops entrance  
↓  
Workstation 04
```

That makes the organisation feel much more believable.

Proposed Alice Prologue route

```
ALICE_INTERNAL_ENTRY  
↓  
Central Ops circulation  
↓  
WS04_AliceMarker  
↓  
speak / USB  
↓  
leave through internal operational route
```

Then later:

```
Alice remembers File 3  
↓  
returns through SAME route  
↓  
WS04
```

This gives continuity.

She does not simply despawn and reappear beside the player.

A6.6 — Alice around Workstation 04

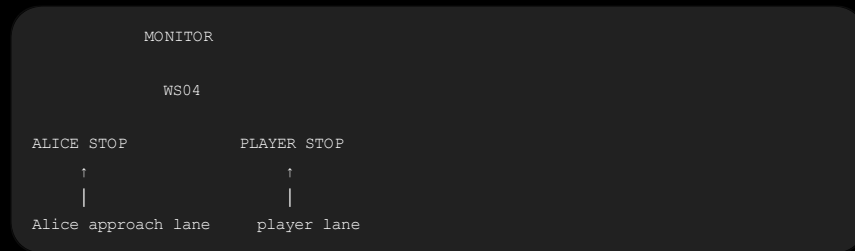
We already have:

```
WS04  
(+4.0, 0, -2.0)  
  
Player marker  
(+4.0, 0, -3.2)  
  
Alice marker  
(+2.8, 0, -2.4)
```

This still works.

But we need an arrival route that avoids the player's interaction position.

Conceptually:

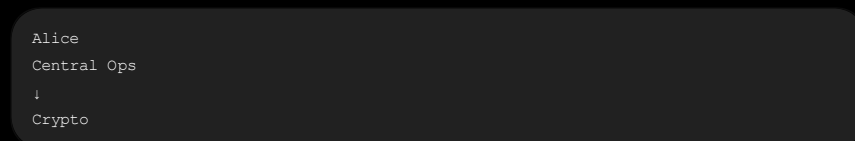


Alice should approach from the **side**, not from directly behind the player.

That will make camera framing substantially better.

A6.7 — Alice in Incident 02

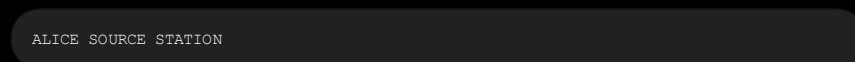
When Cryptography opens:



She may initially accompany or already be present at the Crypto Lab.

But once the learner starts configuring protection, Alice does not need to stand beside every interface.

She can occupy a dedicated:



This reinforces her role:

Alice = source/origin.

A6.8 — Alice in Incident 03

Her main route:



After the symmetric workflow is proven, Alice returns to operational work.

Then the payment request occurs naturally in Central Operations.

That feels much stronger than Alice remaining in the laboratory for the entire mission.

A6.9 — Alice in Incident 04

Alice should remain primarily in Central Operations.

Her importance is:



When Bob reports the unexpected R 1 000 payment, the learner compares Alice's known instruction against Bob's result.

Alice **does not follow the learner into SOC** simply because the investigation moves there.

The evidence moves into SOC.

Alice remains an operational source.

That is a very important distinction.

A6.10 — Alice in Incidents 05–06

She becomes much less physically important.

That is good.

She should not be artificially inserted into:

```
Red Team
target investigation
offensive authorisation
financial recovery
```

Her historical evidence remains relevant.

Her physical presence does not need to.

A6.11 — Bob

Bob should be treated fundamentally differently.

Bob is remote.

His route is not:

```
Bob walks around Headquarters.
```

His route is an **information route**.

```
BOB REMOTE ENDPOINT
|
▼
COMMUNICATIONS
|
|— Crypto interface
|
|— Central Ops workstation
```

A6.12 — Bob's physical representation

We can represent Bob through:

```
video/character communication window
remote terminal
role-status display
3D communication avatar/window later
```

But he should not suddenly materialise physically inside Central Ops.

That maintains the communications concept.

Bob's campaign presence

```
PROLOGUE
Central Ops via remote workstation
```

```
INCIDENT 02
Communications

INCIDENT 03
Communications / Crypto endpoint

INCIDENT 04
Central Ops → Communications evidence

INCIDENT 05
limited operational involvement

INCIDENT 06
not primary
```

Bob's **network presence follows the learner**, not Bob himself.

A6.13 — Eve

Eve requires particular discipline because we do not want her movement to accidentally imply guilt.

Her legitimate functional route should be:

```
IDENTITY / ACCESS
↓
COMMUNICATIONS
↓
SOC
```

This matches authorised security support.

A6.14 — Eve in the Prologue

Eve's first activity occurs around:

```
RECEPTION / IDENTITY
```

because she performs the authorised training credential check.

That makes perfect organisational sense.

She does not need to follow the trainee into Central Operations immediately.

A6.15 — Eve after credential setup

After completing the authorised check:

```
Eve
Identity/Security station
↓
internal operational route
↓
Communications / security support
```

The player continues:

```
Reception
↓
Central Operations
```

Their paths separate.

This is valuable because Eve does not become visually attached to every later anomaly.

A6.16 — Eve during PQ-001

When the suspicious credential event appears, the learner checks:

Is this explained by Eve's authorised activity?

That check can occur digitally.

We do **not** need Eve suddenly appearing beside Workstation 04.

Result:

```
AUTHORISED EVE TEST
known

LATER EVENT
not explained
```

The game keeps the distinction clear.

A6.17 — Eve during SOC incidents

From Incident 04 onward, Eve may legitimately appear within:

```
COMMUNICATIONS
SOC
```

doing security-support work.

That actually helps reinforce that she is **part of the authorised security organisation**.

She should not lurk alone in corners.

She should be working alongside normal SOC personnel.

A6.18 — Eve and Red Team

For Release 1:

```
EVE → RED TEAM
not a normal route
```

Not because she is suspicious.

Simply because her current role does not require it.

That protects organisational separation.

A6.19 — Red Team Unit 01

Red Team has the strongest physical route restriction of our Release 1 characters.

Normal home:

Red Team Operations

Allowed operational relationship:

```
RED TEAM
↔
SOC
```

through T2.

That is basically it for normal campaign movement.

A6.20 — Red Team before Incident 05

Earlier in the campaign:

```
RED TEAM
ACTIVE ORGANISATIONAL UNIT

PLAYER ACCESS
NO
```

The player may see:

- staff silhouettes;
- activity;
- secure displays;
- personnel entering through T2.

This tells the learner:

Phantom Q already has capabilities you have not yet earned access to.

Excellent environmental progression.

A6.21 — Incident 05 Red Team standby

At:

```
RED TEAM STATUS
STANDBY
```

Red Team Unit 01 should visibly begin preparing.

Possible environmental movement:

```
RT operator
workstation
↓
secure console
↓
team briefing point
```

But they remain **inside Red Team**.

The learner stays outside.

The learner can see operational activation without being granted access.

A6.22 — Incident 06 Red Team movement

Once authorised:

```
RED TEAM
↔
SOC
```

becomes a major route.

However, we should avoid making every scene physically move the learner and RT staff together.

Instead:

Investigation block

Some Red Team personnel can move into SOC as liaison/support.

Active operation block

Return to Red Team.

Evidence/recovery block

SOC receives the signed evidence digitally and operational personnel can correlate it there.

That makes the departments feel connected without becoming a parade of NPCs.

A6.23 — HQ Command / Authorisation personnel

This is another group that should **not chase the learner**.

Their home is the restricted internal area of:

Central Operations

```
CENTRAL OPS
|
├─ operational floor
|
└─ Command / Authorisation zone
```

Their normal movement:

```
Command station
↔
operations display
↔
restricted briefing position
```

Not:

```
Command officer
↓
walks to Red Team every time approval is needed
```

That would undermine Incident 05's communication architecture.

A6.24 — Incident 06 authority route

The authority itself moves digitally:

```
HQ COMMAND
  |
HQ private signature
  |
  ▼
SESSION / NETWORK
  |
  ▼
RED TEAM
```

The **person does not need to physically move**.

This is exactly the distinction A6 needs to preserve.

A6.25 — SOC personnel

SOC staff should have a recognisable route cluster:

```
SOC
↔
COMMUNICATIONS
↔
Central Ops liaison point
```

But most remain inside SOC.

Typical movement:

```
analyst workstation
↓
evidence wall
↓
correlation station
↓
briefing table
```

One or two designated analysts can use the Central Ops connection.

Not every SOC employee.

A6.26 — SOC traffic and A4

This exposes something we need to preserve physically.

Because SOC connects to both:

```
Central Ops
Communications
Red Team
```

we must avoid creating one tiny doorway through which all three traffic streams converge.

I recommend SOC having **three distinct edges**:

```
SOC
|
├─ T1A Central Ops portal
|
├─ T1B Communications portal
|
└─ T2 Red Team vestibule
```

This is an A6 refinement to A4.

We do **not** need to move SOC.

But its doorway design must support three functional flows.

A6.27 — Reception / Identity personnel

Their route should remain very local.

```
RECEPTION
IDENTITY DESK
ACCESS GATE
```

They do not casually walk into:

```
Crypto
SOC
Engineering
Red Team
```

during ordinary gameplay.

This strengthens the sense that departments have actual personnel boundaries.

A6.28 — Foundations personnel

Training staff route:

FOUNDATIONS
↔
Central Ops
↔
Engineering occasionally

They can also appear as mentors/instructors later.

They should not normally use:

SOC
Red Team
Secure Core

without specific mission justification.

A6.29 — Cryptography personnel

Home cluster:

CRYPTO
↔
COMMUNICATIONS

Secondary:

CRYPTO
↔
Advanced Compute

through controlled access later.

This gives Crypto staff an actual technical identity.

A6.30 — Communications personnel

Communications has the broadest ordinary operational relationships:

CRYPTO
|
CENTRAL OPS — COMMUNICATIONS — SOC

This confirms our A2/A3 decision that Communications is a bridge department.

Communication staff should be one of the most mobile specialist groups.

A6.31 — Engineering personnel

Their main route:

ENGINEERING
↔
QUANTUM
↔
SERVICE SPINE

Secondary:

ENGINEERING
↔
Foundations / operational technical gate

This is important.

Engineering staff carrying equipment should **not** use:

```
Reception
↓
Central Ops
↓
Crypto
↓
Quantum
```

The service spine exists precisely to avoid that.

A6.32 — Equipment has a route too

This is worth treating almost like a character route.

Large equipment:

```
SERVICE ENTRY
↓
SERVICE SPINE
↓
ENGINEERING
↓
QUANTUM
```

Examples:

- QKD hardware;
- optical equipment;
- server racks;
- electronics;
- test instruments.

This confirms the service route was not decorative.

A6.33 — Quantum personnel

Normal route:

```
QUANTUM
↔
ENGINEERING
↔
Advanced Technical Vestibule
```

Potential advanced route:

```
QUANTUM
↔
Secure Core vestibule
```

only with high clearance.

They do not routinely appear in Central Ops.

That makes encountering specialist Quantum staff later feel meaningful.

A6.34 — Advanced Compute personnel

Natural relationships:

```
ADV COMPUTE
↔
CRYPTO
↔
SOC
```

but mostly through controlled technical connections.

They should not behave like general HQ office staff.

A6.35 — Secure Core personnel

This should be extremely limited.

Rather than generic NPC traffic:

Secure Core
busy corridor
20 people walking around

we want:

few authorised personnel
controlled entry
deliberate movement

The lack of traffic reinforces the security level.

A6.36 — Ambient HQ population

We should not populate the facility randomly.

Use distinct populations.

Operational staff

Central Ops
Communications
Reception

Moderate activity.

Specialist staff

Foundations
Crypto
Engineering

Lower but focused activity.

Security staff

SOC
Red Team

Controlled movement.

Advanced technical

Quantum
Advanced Compute
Secure Core

Low visible density.

This means moving deeper into HQ also changes the **social density**.

That's subtle environmental storytelling.

A6.37 — HQ state should affect NPC movement

One persistent building can feel very different by changing character behaviour.

Normal

routine walking
normal workstation use
relaxed operational density

PQ-001

SOC activity increases
one security liaison enters Central Ops
normal staff still working

Not full panic.

Incident 04

SOC movement increases
Communications staff focused
Central Ops activity becomes restricted
Red Team remains behind boundary

Incident 05

SOC active
Crypto staff active
Communications active
Red Team visibly preparing

Incident 06

SOC focused
Red Team active
ambient unnecessary traffic reduced
Command area active

Restored

normal traffic gradually returns

The building therefore changes through **people**, not only monitor colours.

A6.38 — Character-route priority

Unity navigation later needs priority rules.

Conceptually:

```
PRIORITY 1
Scripted mission character
```

```
PRIORITY 2
Player interaction target
```

```
PRIORITY 3
Operational NPC
```

```
PRIORITY 4
Ambient NPC
```

So if Alice must reach WS04:

```
ambient NPC should not
stand in Alice's destination
```

This sounds small, but it matters enormously for polish.

A6.39 — Door behaviour with NPCs

NPCs must also obey access rules.

We should never have:

```
PLAYER:
ACCESS DENIED

random NPC:
walks through locked door with no response
```

unless the NPC visibly has the required clearance.

So character definitions eventually include:

```
Role
Department
Clearance
PermittedZones
```

Door animation then works for player and NPCs consistently.

A6.40 — Security boundaries become more believable

Example:

Player sees Red Team staff approach T2.

```
Red Team badge
↓
reader validates
↓
vestibule opens
↓
operator enters
↓
door closes
```

Player approaches during Incident 04:

```
TRAINEE CLEARANCE
↓
DENIED
```

Same door.

Same rule.

Later Incident 06:

```
MISSION AUTHORISATION
↓
GRANTED
```

That is excellent world-building.

A6.41 — NPCs should not stand in doorways

We should establish a simple spatial rule:

```
NO IDLE MARKERS
within approximately 1.5 m
of major door thresholds
```

And:

```
NO DIALOGUE MARKERS
directly inside main circulation paths
```

Interaction points need small **activity pockets** beside routes.

This will improve handling dramatically.

A6.42 — Camera-aware NPC routes

Because of our fixed raised-oblique camera, we should avoid constant NPC movement across critical foreground space.

For example, during:

```
Alice → WS04 conversation
```

ambient staff traffic immediately between the camera and Alice should temporarily reduce.

Not freeze the entire world.

Just prioritise the narrative sightline.

A6.43 — Route markers versus fixed paths

We should **not manually author every step**.

Unity eventually gets destination markers:

```
ALICE_INTERNAL_ENTRY
ALICE_WS04
ALICE_CRYPTO_SOURCE
ALICE_COMM_STATION

EVE_IDENTITY
EVE_COMM_SECURITY
EVE_SOC

RT_DESK
RT_BRIEFING
RT_SOC_LIAISON
```

NavMesh handles the physical route.

We only prescribe intermediate waypoints where:

- security gates matter;
- camera composition matters;

- a character must avoid a particular zone.

This keeps movement flexible if furniture changes.

A6.44 — Character home points

Every important NPC should have a logical **home state**.

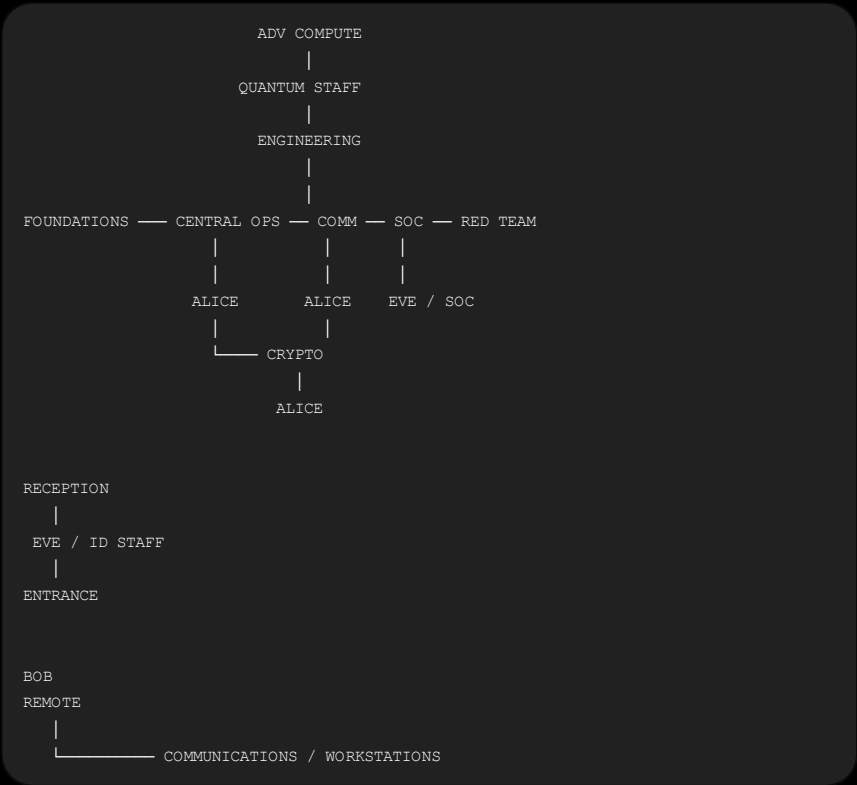
Character	Home location
Alice	Central Ops / operational source station
Bob	Remote endpoint
Eve	Security Support / Communications
Red Team Unit 01	Red Team Operations
HQ Command	Command sub-zone, Central Ops
SOC analysts	SOC
Reception staff	Reception
Crypto staff	Crypto Lab
Engineering staff	Engineering
Quantum staff	Quantum Wing

Then if no mission behaviour exists:

```
NPC
|
returns to home behaviour
```

rather than standing wherever the last cutscene ended.

A6.45 — Major route map by character



The important thing is that there is **no character who needs to wander through every department**.

That tells us the organisational separation is healthy.

A6.46 — Character campaign matrix

Character	Prologue	I01	I02	I03	I04	I05
Player	Reception → Ops	Foundations	Crypto/Comm	Comm/Crypto/Ops	Ops/Comm/SOC	SOC/Crypto/C
Alice	Ops	supporting evidence	Crypto	Crypto → Ops	Ops	limited
Bob	remote	evidence	remote/Comm	remote	remote	limited
Eve	Identity/Security	baseline evidence	support only	support only	Comm/SOC	SOC support
RT Unit 01	restricted/off-screen	restricted	restricted	restricted	restricted	standby
HQ Command	Central Ops	Central Ops	Central Ops	Central Ops	incident oversight	trusted comm
SOC staff	restricted activity	limited	background	background	active	active

This looks coherent.

A6.47 — A6 exposes a real correction to A4

The important correction is not room placement.

It is **internal entry geometry in Central Operations**.

We need at least two meaningful access directions:

SOUTH
Player / Reception entry

NORTH or NORTH-WEST
Internal operational staff / Communications side

This lets Alice and staff enter naturally without walking through the trainee entry sequence.

So A4's Central Operations design should reserve:

T0 SOUTH
Reception / trainee entry

INTERNAL OPS NORTH
staff / Communications circulation

That is a useful refinement.

A6.48 — A6 exposes a second refinement

SOC needs distinct circulation for:

OPS

COMMUNICATIONS

RED TEAM

Therefore the SOC room should not use one central door for everything.

We preserve:

```
SOC PORTAL A
Central Ops

SOC PORTAL B
Communications

T2
Red Team
```

This will make both navigation and storytelling cleaner.

A6.49 — A6 exposes a third refinement

Crypto ↔ Communications will carry frequent:

```
player
Alice
specialist staff
```

movement.

The connection should therefore be wider than an ordinary 1.0 m office door.

I would reserve:

1.8 m controlled double-width opening

or roughly equivalent specialist connection.

Not necessarily a dramatic double door visually.

Just sufficient clear width.

A6.50 — Loop against A1

Does the building still feel like an organisation?

Yes — more so now.

Characters have:

- departments;
- access;
- routes;
- reasons for moving;
- home states.

They are not merely mission props.

A1 passes.

A6.51 — Loop against A2

Does spatial hierarchy still work?

Yes.

Higher-security personnel move increasingly inside restricted branches.

We do not need to flatten the hierarchy for character convenience.

A2 passes.

A6.52 — Loop against A3

Do department relationships support staff movement?

Very well.

Particularly:

```
Alice:
OPS ↔ COMM ↔ CRYPTO

Eve:
ID ↔ COMM ↔ SOC

Red Team:
SOC ↔ RT

Engineering:
ENG ↔ QUANTUM ↔ SERVICE
```

This validates the strongest A3 relationships.

A3 passes.

A6.53 — Loop against A4

The room positions survive.

We only need three refinements:

1. Internal staff entry into Central Ops
2. Three distinct SOC circulation edges
3. Wider Crypto ↔ Communications connection

No major room relocation required.

A4 passes with refinement.

A6.54 — Loop against A5

Character movement does not conflict with campaign movement.

More importantly, characters do **not** need to follow the player through every campaign location.

That means A5 becomes cleaner rather than more complicated.

A5 passes.

A6 — Canonical character-route lock

I would now carry these rules forward:

Alice belongs primarily to Central Operations, Communications and Cryptography; Bob remains remote; Eve belongs to authorised Identity/Security Support, Communications and SOC; Red Team operates primarily between Red Team Operations and SOC; HQ Command remains physically rooted in Central Operations while authority travels digitally; Engineering/Quantum personnel use the technical branch and service spine rather than operational/public circulation.

And for the world itself:

```
NPC MOVEMENT
must reflect role
```

```
SECURITY
must apply to NPCs too

SCRIPTED NPC
takes route priority

AMBIENT STAFF
must not block gameplay

CHARACTERS
have persistent home locations

INCIDENT STATE
changes movement patterns

REMOTE PARTICIPATION
does not require physical teleportation
```

With A6 complete, the Headquarters is now working not only as a campaign map but as an **operating organisation**.

The next loop, **A7 — Clearance Progression**, is where we take this exact floor plan and character structure and determine precisely what the trainee can **see, enter and use** at the Prologue, after PQ-001, through each Incident, and at the end of Incident 06—while separating **permanent capability clearance** from **temporary mission authorisation**.